The particle model of matter (1)

There are three states of matter: solid, liquid and gas. The **particle model** is a scientific explanation that describes the differences between each state. When we imagine particles in the three states of matter, it is as if we are looking at objects with a very powerful microscope and seeing what is happening inside them.

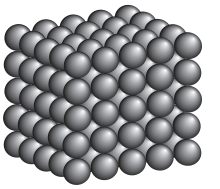
Solids

The particles in solids:

- are very close together
- are tightly held together
- have a small amount of energy
- can only move by vibrating.

Because of this, solids:

- are heavy
- have a definite shape
- do not flow easily
- are hard to compress.



The arrangement of particles in solids

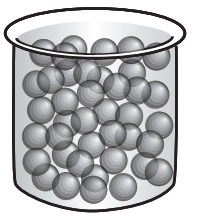
Liquids

The particles in liquids:

- are close together
- are weakly held together
- have a larger amount of energy
- vibrate, but can also slide over each other and change places.

Because of this, liquids:

- are heavy
- can be poured or flow freely
- have no definite shape
- fill the shape of a container
- can't be compressed.



The arrangement of particles in liquids

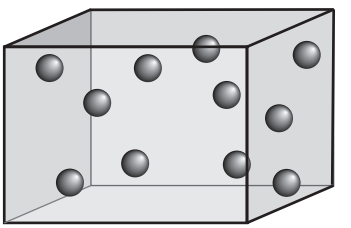
Gases

The particles in gases:

- are spread far apart
- are free from each other
- have a large amount of energy
- move around quickly and freely in all directions.

Because of this, gases:

- are light
- can change shape
- flow freely
- do not have a definite shape or volume
- spread out to fill a container
- can be compressed.



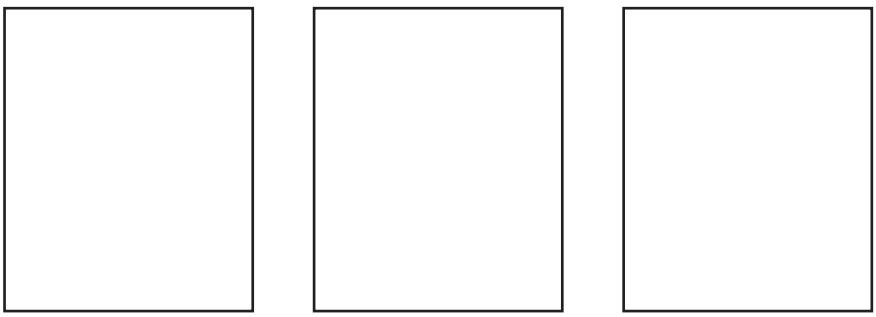
The arrangement of particles in gases

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The particle model of matter (2)

☞ Read the Resource sheet 'The particle model of matter (1)', then answer these questions.

1 In the boxes below, show how particles are arranged in each state of matter.



Solids Liquids Gases

2 Use the phrases in the word bank to complete the table below. Some phrases may be used more than once.

Word bank

slide over each other flow freely are heavy fill the shape of a container
are light have a definite shape can change shape don't flow easily
only vibrate move quickly can't be compressed

The particles in solids ...	**The particles in liquids ...**	**The particles in gases ...**
Because of this, solids ...	**Because of this, liquids ...**	**Because of this, gases ...**

3 Decide whether each of these statements is true or false.

a Liquids have a fixed shape. _____

b Gases are light. _____

c Particles in solids can move freely. _____

d Liquids flow freely. _____

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